

SECTION 4: Image Generation AI

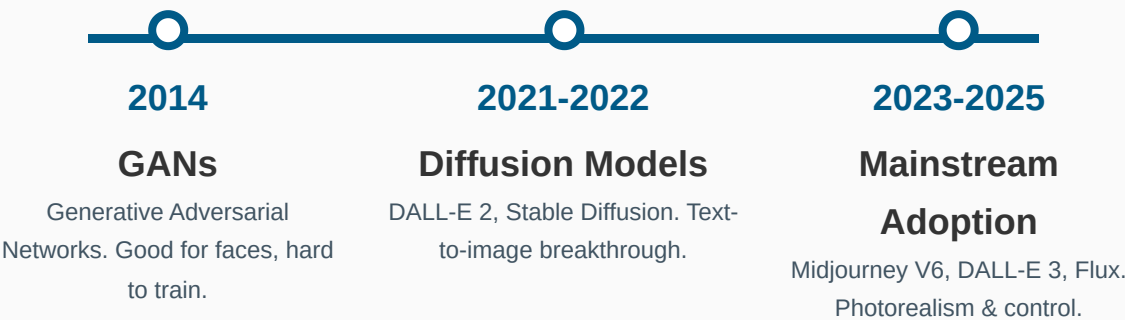
In this section, we transition from text to vision. You will learn the mechanics behind tools like DALL-E, Midjourney, and Stable Diffusion. We cover the evolution from GANs to Diffusion models, how text prompts are converted into pixels, and practical strategies for prompting.

Lecture 4.1: Introduction to Image Generation

The Creative Revolution

Image generation has evolved from simple, blurry experiments to photorealistic, high-resolution art in just a few years. This shift creates images from scratch based on learned patterns, rather than just retrieving existing ones.

Figure 1: Evolution Timeline



Major Models Landscape

Model	Developer	Key Strengths
DALL-E 3	OpenAI	Excellent prompt adherence, text rendering, ChatGPT integration.
Midjourney	Midjourney Inc.	Superior artistic style, lighting, and aesthetics.
Stable Diffusion	Stability AI	Open-source, highly customizable, run locally.

Flux	Black Forest Labs	State-of-the-art quality (2025), high prompt fidelity.
Adobe Firefly	Adobe	Commercially safe (trained on licensed stock), tool integration.

Key Capabilities

- **Text-to-Image:** Creating images from descriptions.
- **Image-to-Image:** Transforming existing images (e.g., sketch to photo).
- **Inpainting:** Editing specific parts inside an image.
- **Outpainting:** Extending an image beyond its borders.
- **Upscaling:** Increasing resolution while adding detail.

LECTURE 4.1 KEY TAKEAWAY

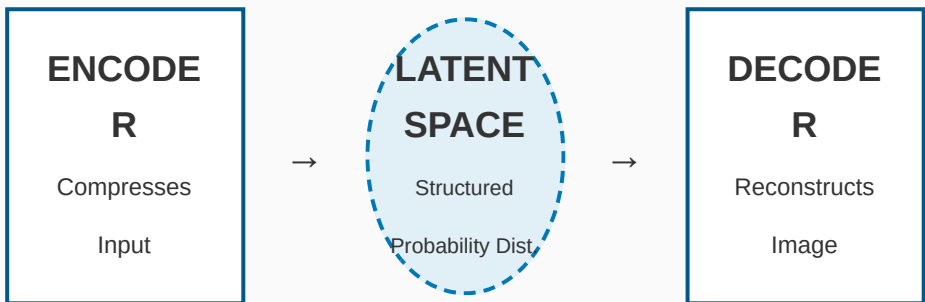
Image generation has moved from research labs to daily creative tools due to breakthroughs in Diffusion Models and massive scale training. While powerful, they still struggle with consistency, hands, and complex spatial relationships.

Lecture 4.2: Variational Autoencoders (VAEs) Explained

The Concept of Latent Space

VAEs are crucial for understanding how we compress visuals. They don't just compress data; they organize it into a **continuous, structured latent space**.

Figure 2: VAE Architecture



Instead of pixel-perfect copies, VAEs learn the "recipe" (mean & standard deviation) to recreate the image.

Why VAEs Matter for Stable Diffusion

Stable Diffusion is a **Latent Diffusion Model**. It doesn't process raw pixels (which is slow); it uses a VAE to compress the image by 8x (e.g., 512x512 → 64x64 latent).

- **Efficiency:** Diffusion happens in the small latent space.
- **Interpolation:** You can smoothly transition between two images by walking through the latent space.

Latent Space Interpolation

If Point A is "Smiling Face" and Point B is "Frowning Face", the midpoint in a VAE latent space isn't a messy overlay—it's a **Neutral Face**. The space captures semantic meaning.

LECTURE 4.2 KEY TAKEAWAY

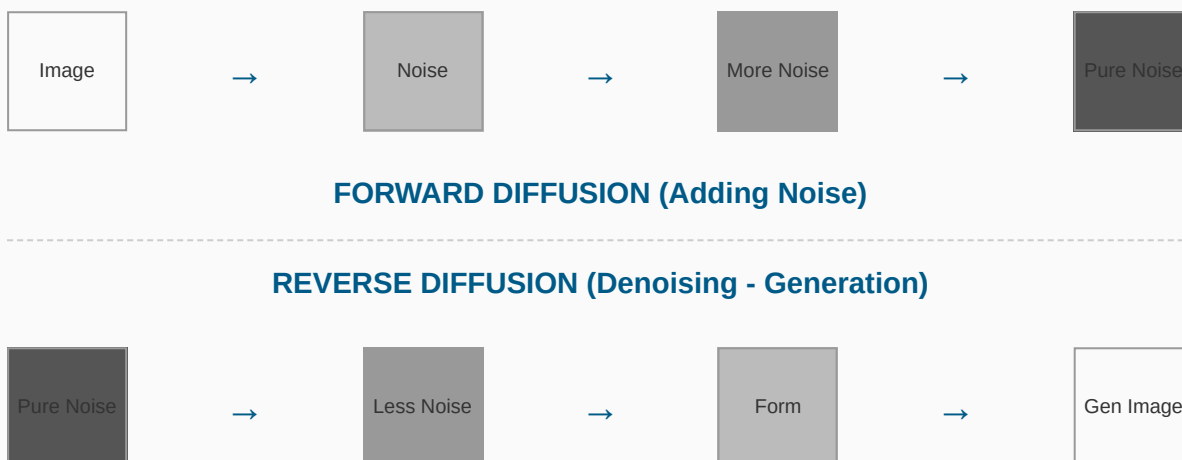
VAEs compress images into mathematical representations (latent codes) that capture meaning. This allows for efficient processing and smooth transitions between concepts, serving as the backbone for modern Latent Diffusion Models.

Lecture 4.3: Understanding Diffusion Models

The Core Mechanism: Denoising

Diffusion models are based on a counterintuitive idea: **learning to reverse destruction**.

Figure 3: The Diffusion Process



How It Works

1. **Forward Process:** Gradually add Gaussian noise to an image until it's random static. (Deterministic)
2. **Training:** Teach a neural network (U-Net) to predict the noise added at each step.
3. **Generation (Reverse):** Start with pure random noise. Ask the model "What noise is here?" and subtract it. Repeat for 20-50 steps to reveal an image.

Key Components

- **U-Net Architecture:** A neural network shape that processes image features at different scales (coarse to fine).
- **Cross-Attention:** The mechanism that injects your text prompt into the denoising process to guide the outcome.
- **Guidance Scale (CFG):** Controls how strictly the model follows your prompt vs. creative freedom.

LECTURE 4.3 KEY TAKEAWAY

Diffusion models generate images by iteratively refining random noise. They "sculpt" the noise into a coherent image, guided by your text prompt at every step. This iterative process allows for high detail and stability compared to GANs.

Lecture 4.4: How Text-to-Image Generation Works

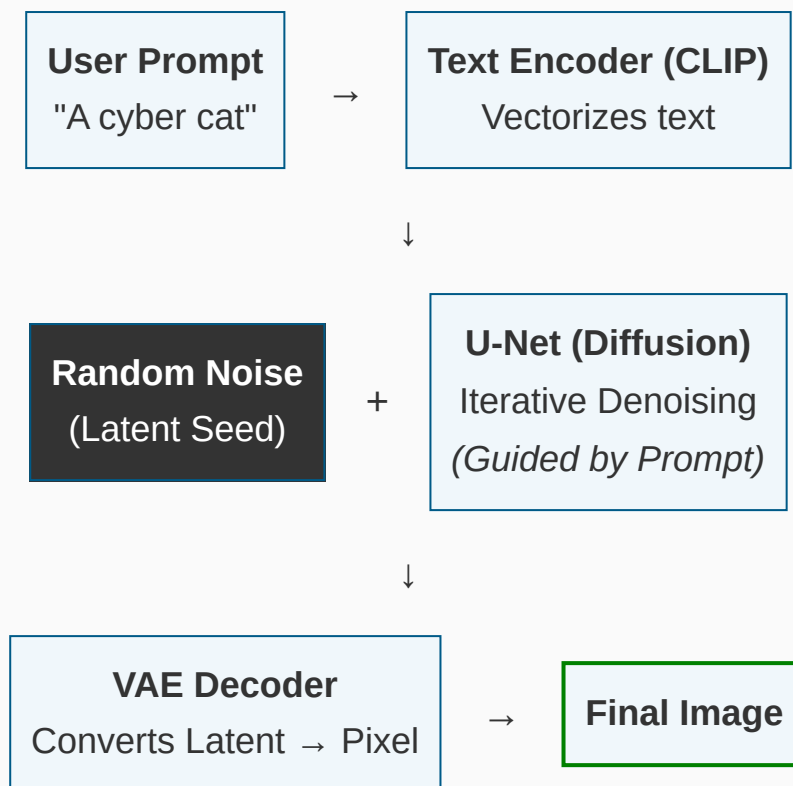
Connecting Language and Vision (CLIP)

To generate an image from text, the computer must understand how words relate to visuals. This is done using **CLIP** (Contrastive Language-Image Pre-training).

The Shared Embedding Space

CLIP is trained to place the text "A dog playing fetch" and an *image* of a dog playing fetch into the same mathematical location (vector) in embedding space.

Figure 4: Text-to-Image Pipeline



Handling Concepts via Cross-Attention

The model uses **Cross-Attention** maps to decide *where* to put things. When generating "A bear in a forest":

- It attends to "Bear" pixels for the center shape.
- It attends to "Forest" pixels for the background.

Improvements in 2025

- **Text Rendering:** Models like Ideogram and DALL-E 3 can now spell correctly within images.
- **Consistency:** Tools like ControlNet and LoRA allow consistent characters across different images.

LECTURE 4.4 KEY TAKEAWAY

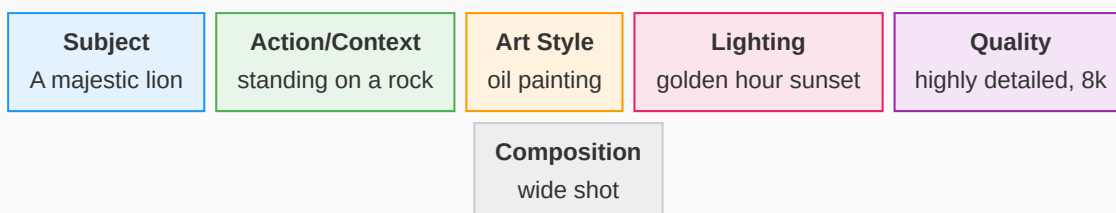
The pipeline combines a Text Encoder (to understand the prompt), a Diffusion Model (to generate the latent structure), and a VAE Decoder (to make it visible pixels). Cross-attention ensures the generated pixels match the specific words in your prompt.

Lecture 4.5: Prompting for Image Generation

The Anatomy of a Perfect Prompt

Unlike chat prompts, image prompts often need to be descriptive "word salads" to trigger specific visual styles.

Figure 5: Prompt Anatomy



Key Modifiers Checklist

- **Subject:** Be specific (e.g., "Tabby cat" instead of "cat").
- **Medium:** Photograph, 3D render, Sketch, Oil Painting.
- **Style:** Cyberpunk, Steampunk, Minimalist, Baroque.
- **Lighting:** Cinematic lighting, Volumetric lighting (god rays), Soft light.
- **Negative Prompts:** What to remove (e.g., "blurry, bad anatomy, text, watermark").

Iteration Workflow

Generate (4 variations) → **Evaluate** (pick best seed) → **Refine** (add style keywords) → **Upscale**.

LECTURE 4.5 KEY TAKEAWAY

Effective prompting requires specificity. Define the Subject, Medium, Style, and Lighting explicitly. Use Negative Prompts to clean up artifacts. Iterate using seeds to refine a specific look.

Lecture 4.6: GANs - An Alternative Approach

The Adversarial Game

Before Diffusion took over, **GANs (Generative Adversarial Networks)** ruled. They work like a forger and a detective playing a game.

